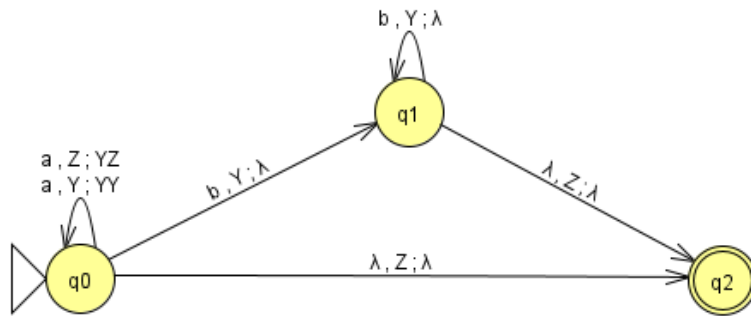


Grammar and parsing

In a context-free grammar, the left side of a rule is always a single nonterminal. This makes it context-free, as what is on the left or right side of the nonterminal does not affect the choice of the rule.


 $a^n b^n$

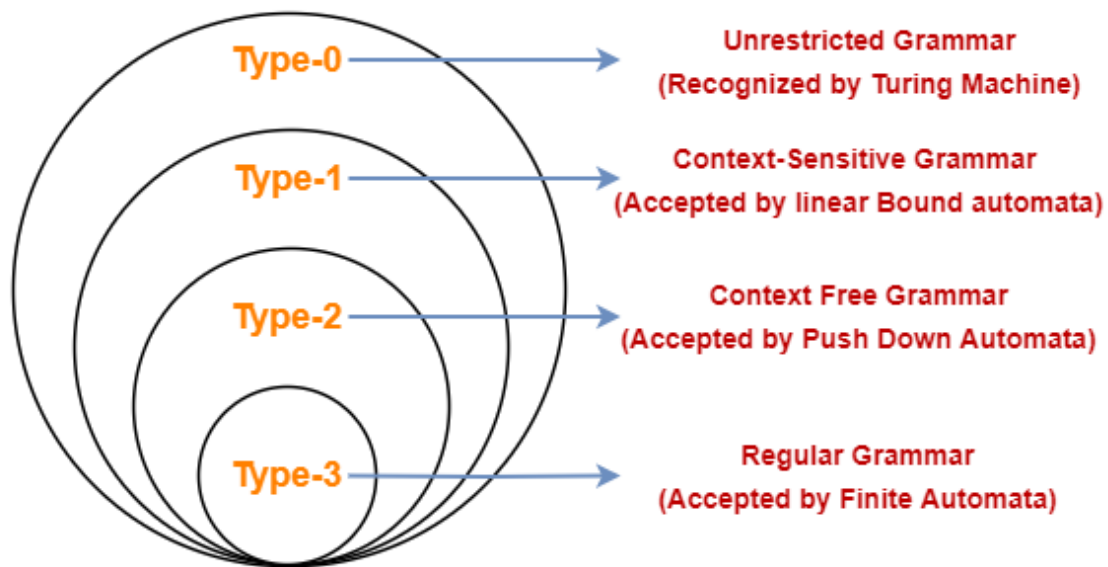
LHS		RHS
S	$\rightarrow$	aAB
S	$\rightarrow$	λ
G	$\rightarrow$	b
A	$\rightarrow$	aAG
A	$\rightarrow$	b
B	$\rightarrow$	λ

A pushdown automaton accepting $a^n b^n$ words produces the above grammar

aaabbb

$S \rightarrow aAB \rightarrow aaAGB \rightarrow aaaAGGB \rightarrow aaabGGB \rightarrow aaabbGB \rightarrow aaabbbbB \rightarrow aaabbbb$

Grammars – Type 1 and 0 languages



Type 1	Context-sensitive	LBA, Linearly bounded Nondeterministic Turing machine	$\alpha A \beta \rightarrow \alpha \gamma \beta$	$L = \{a^n b^n c^n \mid n > 0\}$
Type 0	Unrestricted grammar, Recursively enumerable	Turing machine (infinite tape)	$\gamma \rightarrow \alpha$	$L = \{w \mid w \text{ describes a finite turing machine}\}$

Context-sensitivity: $\alpha A \beta \rightarrow \alpha \gamma \beta$, for the nonterminal A to be used in parsing, it must have α on its left and β on its right, which binds it to the surrounding context.

Both type 1 and type 0 languages are recognized by a Turing machine.
For example $a^n b^n c^n$, $n \geq 1$ which is not context-free ($n \geq 0$, add rule $s \rightarrow \epsilon$)

$S \rightarrow abc \mid aAbc$
 $Ab \rightarrow bA$
 $Ac \rightarrow Bbcc$
 $bB \rightarrow Bb$
 $aB \rightarrow aa \mid aaA$

We will come back to this in lecture 10

Turing machine

A Turing machine is defined (Hopcroft & Ullman 1979, p. 148)

$M = (Q, \Gamma, b, \Sigma, \delta, s(=q_0), F)$

$Q = \{q_1, q_2, \dots, q_n\}$ a **finite** set of states

$\Gamma = \{a, b, c, d, \dots, z\}$ a **finite** set of tape alphabets, can be any symbols

$b \in \Gamma$ is a blank symbol, in JFLAP a blank square

$\Sigma = \{a, b, c, d, \dots, z\}$ a **finite** set of input alphabets (input tape alphabets), can be any symbols

$\delta = (Q \setminus F) \times \Gamma \rightarrow Q \times \Gamma \times \{L, R, S\}$ a **finite** set of state transitions

$(Q \setminus F)$ states Q , excluding final states F

- Γ Tape alphabet from symbols Γ
- Move to one of the states Q
- Write some symbol from Γ on the tape
- The read/write head moves either right (R), left (L), or stays in place (S)

$s(q_0) \in Q$ initial state

$F \subset Q$ (non-empty) set of final states

Alan Turing

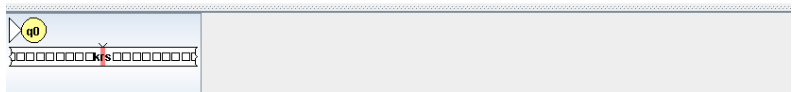
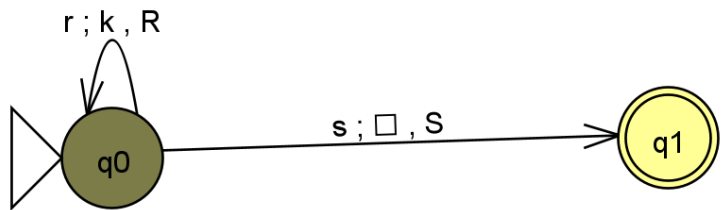
OBE FRS



Turing in 1951

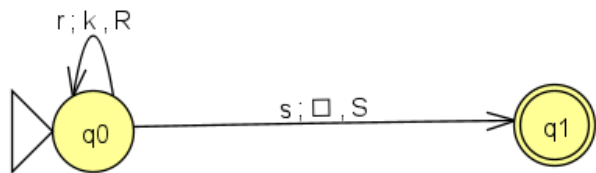
Born	Alan Mathison Turing 23 June 1912 Maida Vale , London, England
Died	7 June 1954 (aged 41) Wilmslow , Cheshire, England
Cause of death	Cyanide poisoning as an act of suicide ^[note 1]
Alma mater	University of Cambridge (BA, MA) Princeton University (PhD)
Known for	Cryptanalysis of the Enigma · Turing's proof · Turing machine · Turing test · unorganised machine · Turing pattern · Turing reduction · "The Chemical Basis of Morphogenesis" · Turing paradox
Awards	Smith's Prize (1936) Scientific career
Fields	Logic · mathematics · cryptanalysis · computer science · mathematical and theoretical biology ^[1]

Simple Turing machine



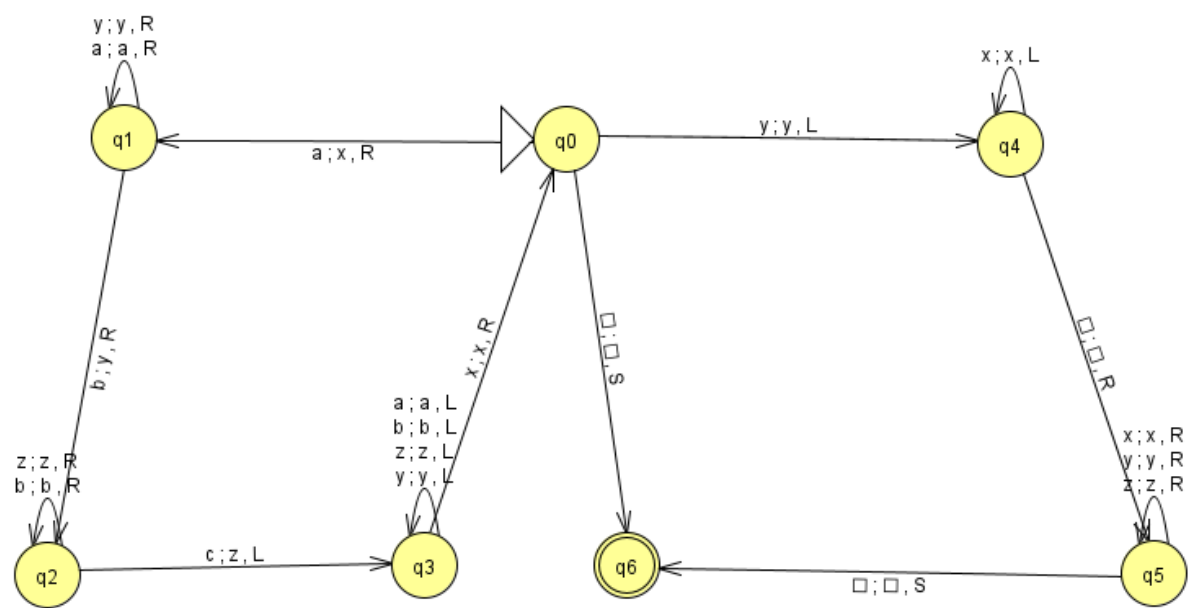
Transition: read;write;transition(L/R/S)

recognizes words r^*s (does not consider the symbols on the tape after this)



Input	Result
rrs	Accept
rs	Accept
rrrrrsss	Accept
rsss	Accept
r	Reject
ssss	Accept
s	Accept

$a^n b^n c^n$



Palindrome as a Turing machine

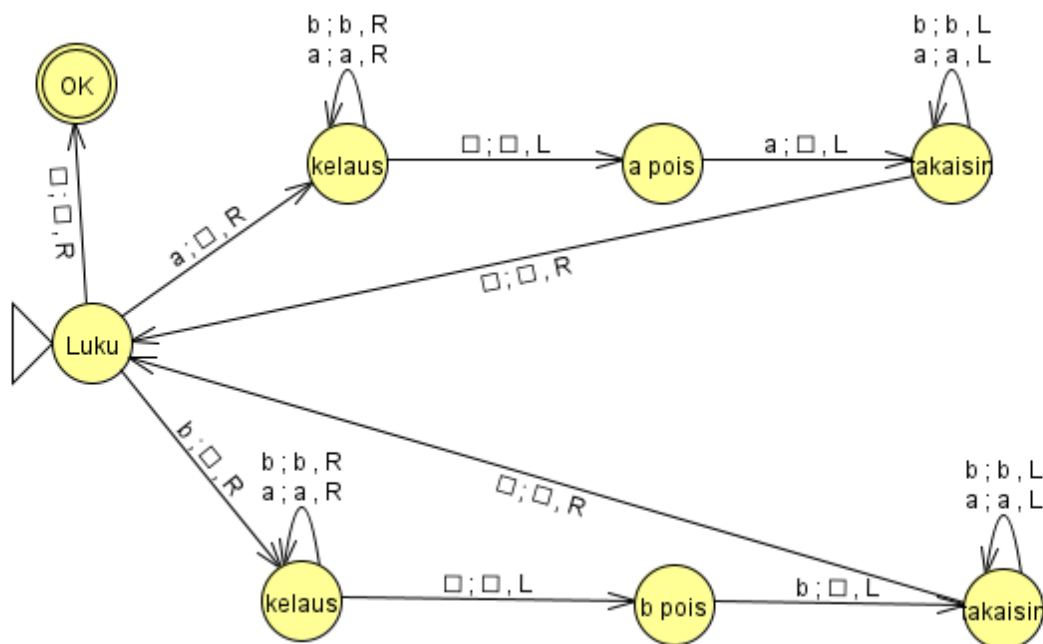
Principle:

- 1 Read s,a i p,u or k erase it _____saippuakauppias_____
- 2 Go to the end of the tape and erase the same symbol there _____saippuakauppias_____ resulting in _____aippuakauppia_____
- 3 Return to the beginning and repeat until the letters run out after the last erasure

BTW: Saippuakauppias means a soap dealer

Palindrome Turing, alphabet a,b

The following deals with palindrome in the alphabet a,b e.g. ___abba___



Time requirement?

n characters

n erasures

n passes over the word

→ $O(n^2)$